Cold active-support certificate preservation

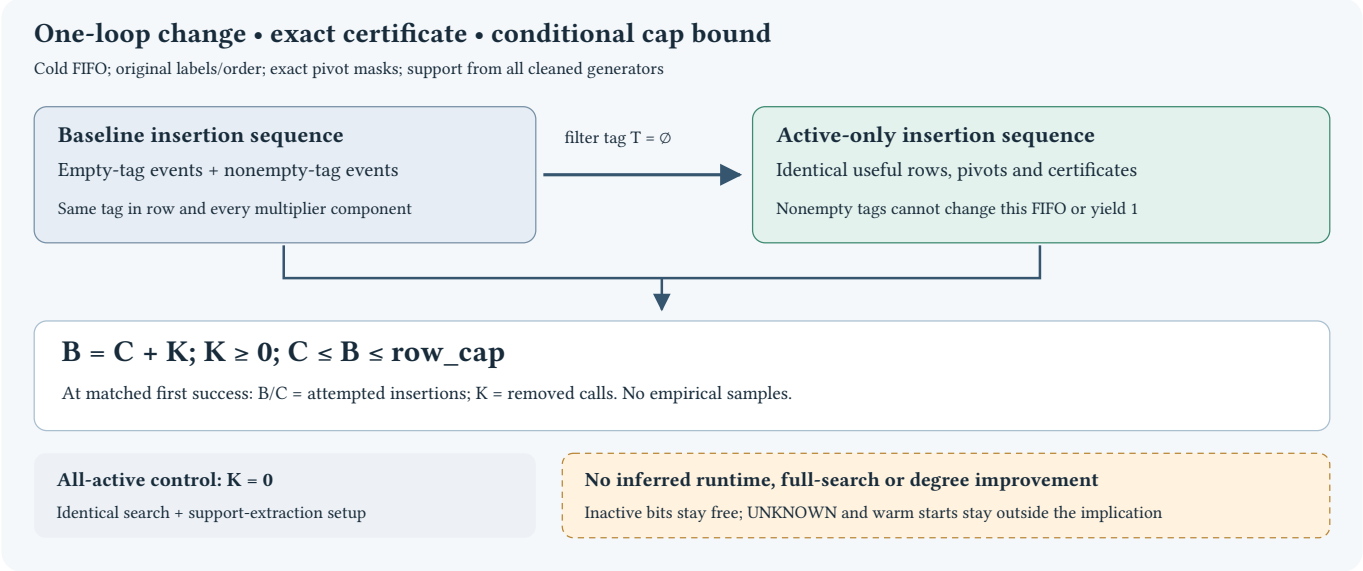
DEC-20261030-722a2d · IDEA-20261030-b6f05e · 30 October 2026

Approve scoped archival acceptance of the exact cold source implication. When the bound baseline refutes, changing only its multiplication loop to increasing original active-variable indices reproduces the **identical ordered multiplier certificate**, with no more attempted insertions and within the same row cap. One independent reviewer completed two sealed phases; all seven joints hold and the unchanged native gate passed. This is a manual source derivation, with zero new runs.

Exact premises and proof

Over a prime field, clean every ordered generator modulo the prime. Support A includes every variable in every cleaned generator, even generators above the degree cap. Preserve original labels, generator and active-index order, exact leading-mask order, cold empty basis, FIFO, normalization, limits and stopping. Attempts count **before** cleaning and degree rejection, including zero, dependent and over-degree calls. Valid masks and primality are mathematical premises.

Every row and each multiplier component carry the same exact inactive mask T. Exact pivot equality prevents reductions across T. Nonempty-tag rows cannot change the empty-tag basis or yield a constant. Filtering baseline events and FIFO to the empty tag therefore gives the literal candidate trace. Prefix injection preserves the common cap through the identical first certificate.



Symbolic proof diagram; not to scale and not an empirical graph. B and C count attempted insertions up to matched success. K counts removed calls. No measured gain or uncertainty estimate is asserted.

Seven reviewed joints

Joint	Scoped conclusion: holds
Algebraic identity	Specialization/lifting preserves certificates; inactive model bits stay free.
Source invariant	Rows and all multiplier components preserve a common inactive tag.
Ordered simulation	Empty-tag FIFO reproduces the exact certificate in the original order.
Cap semantics	C ≤ B on baseline success; literal pre-clean insertion counting retained.
Negative controls	All-active overhead, empty support and invalid transfers are worked.
Models and costs	No complete-PC, whole-SAT, degree or total-cost inference.
Producer comparison	Notes agree after reveal; sealed blind proof unchanged.

Source: research/prime_sat_five_tests_20261029/core.py, SHA-256 c67fc44404b282ac963f1021a4b6197ebbb4a8bf382c70a2576db45172065a4f. Detailed derivation and comparison: research/prime_sat_active_support_review_20261030/TASK-20261030-94b46f/.

Boundaries and concrete continuation

Controls distinguish the theorem from stronger claims

All-active support gives K = 0 and adds support-extraction work. Empty support, constants and zero caps retain the ordinary stopping rules. Inactive bits stay free: input x with inactive y permits both (0,0) and (0,1).

The reviewer gives a handwritten order control over any odd prime: f = 1+x+y, g = xf−1 and h = yf−1, with degree cap two. Reversing which active variable is multiplied first changes the first certificate from (x,−1,0) to (y,0,−1), at the fourth insertion. This was not executed. A mixed warm start (1+y)x and divisibility-based pivots break premises of the proof. Alternative support from accepted initial rows would need a separate span argument; it is not universally declared invalid.

Queue exhaustion is UNKNOWN, not complete degree-bounded polynomial calculus or a non-refutability proof. No strict insertion saving or changed capped outcome is exhibited here. Ranks, arithmetic totals and whole-SAT trajectories may differ. Support extraction, construction, restriction, proof production and checking, reconstruction, failures, memory, serialization and I/O remain chargeable. Parent reuse and bookkeeping are separate factors.

Review quality and custody

Blind derivation was sealed before the producer YAML/notes were released. Producer comparison remained inconclusive in that immutable phase and became holds only in the final comparison; no blind mathematical correction was required. The unchanged native gate passed at 18:06:52 UTC. Coordinator rehashed all seven review artifacts and all nine bound source/checker inputs successfully, without rerunning the checker.

This is one reviewer in two phases. Native checks establish byte/declared-chronology consistency, not authenticated timestamps, absence of undisclosed reads or provider identity. The prospectively admitted native request was gpt-6.1-sol/xhigh/fork-none; actual serving model and effective effort are unknown. Policy-output truncation/repeated reads and second-resolution pre-command observations are disclosed. No formal proof, novelty finding, runtime result, new regularity theorem or scalar recovery follows.

Preserve earlier progress; test this factor next

The earlier raw even-s grouped-presentation result remains accepted under its own hypotheses: d_reg changes from 1+3s/2 in **2s variables** to s+1 in **3s variables**. Its sourced mutant-closure upper bound is not a strict actual solving-degree or runtime comparison (DEC-20261029-0857a9). This local source review changes no frontier bound, lead disposition or H/EXP/EV/KN/goal status. The checked canonical source rows remain unchanged; no corpus promotion is made.

Next design: implement only the reviewed cold loop change and its correctness controls, then freeze a paired residual/whole-search cost comparison retaining **quartic, membership, quadratic and quadratic_aa**, A/A, controls, complementary-point reference, complete projected sets and signed rational witnesses. Charge all cost phases; preserve censoring. Keep warm reuse/bookkeeping separate. A breaking implementation trace must be retained; if correctness holds without useful cost improvement, retain that bounded outcome and advance the separate shared-chart lane. This recommendation admits no trial or unchanged scaling.

Complete shared-coefficient signed charts and interval idempotent membership remain that separate lane. Reducing modulo variable U couples coefficients, so idempotent notation alone establishes no quadratic-all-unknowns or degree bound. The general prime-field ECDLP objective stays open.

Full receipts, hashes and limitations: report-inputs.json; detailed decision: ledger/decisions/DEC-20261030-722a2d.yaml. Scoped merge hygiene passed 22 paths; full local ledger validation refused 45,163 off-disk paths without override. A continuation checkpoint passed PUT/fresh-GET equality but excluded these unfinished outputs; restore untested. Root-owned PDF inspection, exact 23-path archive and final publication remain pending at composition.